



1.0	2025/08/19	Release of Report	TLP				
0.1	2025/08/13	Initial release of report	TLP				
Rev.	Issued date	Description	Made by				

Draga AS

Project:

DISC DEXPI Continuation Project

Document title:	Rev:	Page:
DISC DEXPI Gap Analysis	1.0	1 of 15

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TERMS AND ABBREVIATIONS

Term	Explanation
DISC	Digitalization Industrialization Standardization Collaboration; initiative between Equinor, AkerBP, Aibel & AkerSolutions.
PCA	POSC Caesar Association
P&ID	Piping and Instrumentation Diagram
DEXPI	DEXPI: Data Exchange in the Process Industry
NCS	Norwegian Continental Shelf
CAD	Computer Aided Design
NOAKA	North of Alvheim Krafla Askja
Draw.io	Free and open-source diagramming tool
pnb	pnb plants & bytes: software and consulting company involved in DEXPI group.
round-trip	Export from System A, import into System B, modify in System B, export from System B, import back into System A with changes

1 ABSTRACT

The DEXPI initiative, launched in 2011, aims to establish a common data exchange standard for the process industry. Historically, most DEXPI-related work is conducted within standalone projects, with limited focus on transferring DEXPI P&ID files between different CAD vendors.

Prior to the current DISC DEXPI Continuation Project, the NOAKA DEXPI Project documents most of the requirements for DISC partners to implement DEXPI in a standardized way and addressed several gaps between the DISC requirements and the existing DEXPI specification.

Building on this foundation, the current project aims to close the remaining gaps and address integration challenges between CAD vendors, with the ultimate objective of enabling full round-trip transfers of DEXPI P&IDs. A number of these gaps can only be addressed now that the new DEXPI serialization format is approved, moving away from the Proteus schema. As a result, the solutions developed in this project and presented to the DEXPI content team for integration into the DEXPI model will be serializable exclusively using the new format.

This document outlines the outstanding gaps identified and describes how they will be addressed in the continuation project.

In the context of this document, 'gaps' refer to differences between the NCS P&ID engineering modelling requirements and the DEXPI 2.0 standard, and modelling requirements that have been identified but are not yet documented in the requirements specification or incorporated into the blueprint example files.

2 INTRODUCTION

At the conclusion of the NOAKA/DISC DEXPI project, several engineering elements remained either undocumented in the DISC DEXPI Requirements document and/or unmodelled. In addition, further discussions with DISC members revealed further gaps.

Each of these gaps will be addressed in this project and where relevant included in extensions of the blueprint files made available in the earlier project.

The blueprint files will be provided in both the old Proteus schema and the new DEXPI XML format where possible.

Some solutions will only be feasible using the new DEXPI 2.0 serialization, where there exist limitations due to the Proteus schema.

3 IDENTIFIED GAPS

The following gaps have been identified as either carry-on will work from the previous DISC DEXPI project, gaps resulting from decision made during the previous project or as a direct result of implementing the previous modelling and requirements definitions.

3.1 Instrumentation Modelling

In the NOAKA/DISC DEXPI project, a simplified and pragmatic approach was adopted for modelling instrumentation and automation elements, avoiding the use of the full DEXPI instrumentation models for `ProcessSignalGeneratingFunction` / `MeasuringSystem` and `ActuatingElectricalFunction` / `ActuatingElectricalSystem` with their associated sub-component class types. The DISC project's

simplified approach classified all elements within these modelling branches under the `ProcessInstrumentationFunction` class type, with differentiation achieved through type attributes.

This approach was identified as a gap in the previous project, where the expectation was to resolve and remodel the DEXPI instrumentation information model to support NCS P&ID drafting for instrumentation and automation, using discrete class definitions for the various elements.

During the period between the original DISC DEXPI project and this continuation project, significant effort was invested in discussing, drafting, and revising modelling solutions to support NCS P&ID instrumentation and automation drafting. In parallel, the simplified solution was implemented in practice. Implementation results demonstrated that the simplified approach could deliver the desired connectivity and graph information.

Note: The simplified approach is also more easily applied to legacy DEXPI drawing conversion from image or unintelligent CAD tools.

Gap: This 'gap' can be considered closed.

3.2 Vessel/Tank Internal Components

In the DISC project, a decision was made to exclude vessel internals from the modelling and requirements definition, as these components are typically detailed in documents such as the General Arrangement drawing. It has since been determined that vessel/tank internal components are an essential part of the P&ID and must be included in the DEXPI transfer. Consequently, these elements will be incorporated into the model and requirements document for the continuation project.

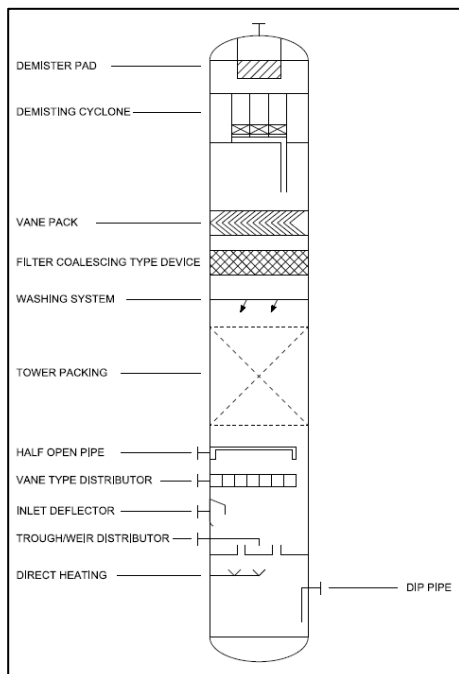


Figure : Example of Vessel/Tank Internal Components.

Gap:

- Document DEXPI suggestion and present to DEXPI Content Team.
- Include updates in DISC DEXPI Requirements document

- Include updates to DISC DEXPI Profile
- Include example in Blueprint files

3.3 Annotations

Annotations (Note) class DISC proposal is part of the DEXPI standard but has not been included in any of the blueprint files for implementation testing.

This class supports note reference, text and association to other DEXPI conceptual objects.

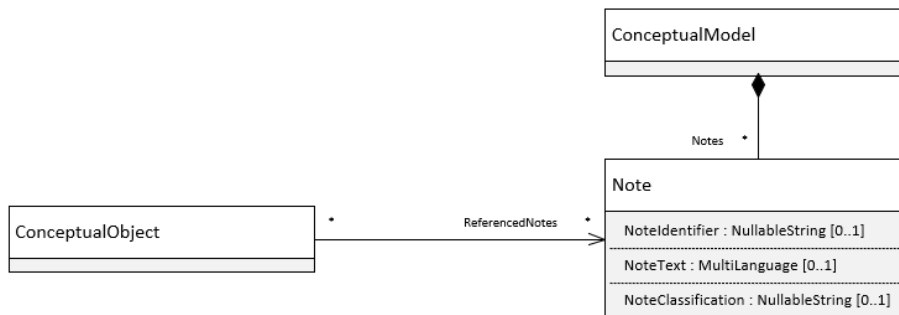


Figure : DEXPI Note class

Gap:

- Include example in Blueprint files

3.4 Mount Type

Mount class DISC proposal recently approved for addition to DEXPI specification.

This class supports the mounting connection between instrument and equipment type and can be assigned as a sensing location for the ProcessSignalGeneratingFunction.

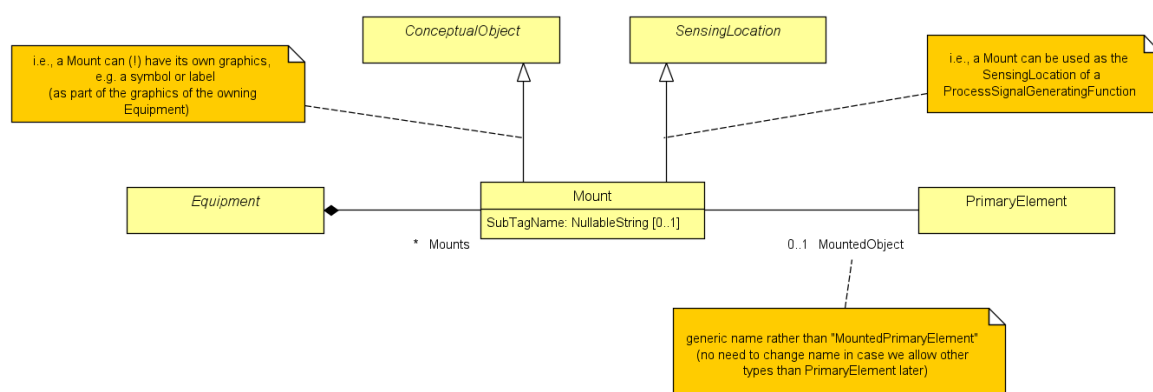


Figure : DEXPI Mount class

Gap:

- Include updates in DISC DEXPI Requirements document
- Include updates in DISC DEXPI Profile
- Include example in Blueprint files

3.5 Report duplicate/incorrect names and URIs

The following items have been identified with duplicate/incorrect names/URIs.

	ComponentClass	ComponentClassURI	note
Fan	BlowerFan	http://sandbox.dexpi.org/rdl/BlowerFan	These two classes are the same except that they have different subtypes. Recommendation is to make a unified class and update documentation or separate them into two classes.
Blower	BlowerFan	http://sandbox.dexpi.org/rdl/BlowerFan	
Bar absolute	Bar	http://data.posccaesar.org/rdl/RDS1314539	All the PressureAbsoluteUnits and PressureGaugeUnits have the same name and URI. The gauge / absolute needs to be always specified to avoid confusion. Also, it is very likely to use both gauge and absolute pressures at the same P&ID sheet.
Bar gauge	Bar	http://data.posccaesar.org/rdl/RDS1314539	

Gap:

- Duplicate/incorrect names Fan/Blower is an error in DEXPI 1.4 – it is fixed in DEXPI 2.0
- Check and if required raise issue with DEXPI group for UoM usage of bar/barg

3.6 Off-Page Signal Modelling

Current DEXPI conceptual model and serialization do not support the transfer of signal lines that run across multiple P&IDs. In these cases, the current standard requires the use of ‘dummy’ elements to ensure the correct compositional topology in the transfer file – fix for orphan signal line. Any solution to this problem must ensure the connectivity information of the signal can be captured and transferred without the use of ‘dummy’ elements.

Gap:

- Discuss and agree on suitable DEXPI model
- Document DEXPI suggestion and present to DEXPI Content Team.
- Include updates in DISC DEXPI Profile
- Include update in DISC DEXPI Requirements document
- Include example in Blueprint files

3.7 Battery Limits and Tie-In Points

Battery Limits define the boundary between plant owner responsibility and external entity responsibility. Tie-In Points defines the boundary between new and existing systems, defining the scope of a project.

Transfer of Battery Limits and Tie-In Points is identified as a gap. These designated location points/group identifiers are not currently part of the DEXPI model.

Gap:

- Discuss and agree on suitable DEXPI model
- Document DEXPI suggestion and present to DEXPI Content Team.
- Include updates in DISC DEXPI Profile
- Include update in DISC DEXPI Requirements document
- Include example in Blueprint files

3.8 ProcessSafetyFunction Subtype

ProcessSafetyFunction as a discrete subtype of ProcessInstrumentationFunction allows for improved modelling for safety critical elements directly from the DEXPI file/model. The expectation is that this new subtype shall be identified as a separate class type.

- Document DEXPI suggestion and present to DEXPI Content Team.
- Include updates in DISC DEXPI Profile
- Include update in DISC DEXPI Requirements document
- Include example in Blueprint files

3.9 Electrical Line Type

Electrical Line additional type to support the electrical lines that are drafted on NCS P&IDs. Within the current DISC DEXPI requirements/profile definition these are modelled as SignalConveyingFunction type. The expectation is that these electrical lines shall be identified as a separate class type e.g. ElectricalLineFunction.

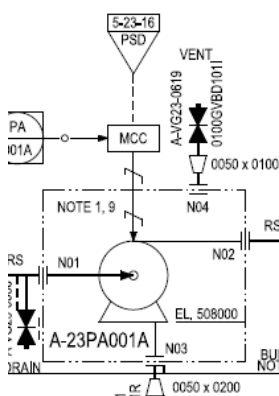


Figure : Example of electrical line between MCC and pump

Gap:

- Discuss and agree on suitable DEXPI model
- Document DEXPI suggestion and present to DEXPI Content Team.
- Include updates in DISC DEXPI Profile

- Include update in DISC DEXPI Requirements document
- Include example in Blueprint files

3.10 Clamp-On

Clamp-on measuring elements are measuring instruments that are fitted externally to the pipe using clamp-on mounting. These measuring devices are not in-line as they are not in direct contact with the process fluid, however, they are often drafted as in-line elements. The current DISC DEXPI requirements models these elements as in-line. This model has been used in projects and is deemed an acceptable solution as these elements are drafted as in-line.

Gap: This 'gap' can be considered closed.

3.11 Motor – Equipment Linking

In DEXPI 1.4, a new concept called TransmissionSystem has been introduced.

This provides a formal way to represent the connection between a motor and its associated equipment (e.g., motor and pump). The model will be incorporated into the DISC DEXPI requirements and will replace the current practice of referencing the equipment ID directly within the motor's data.

Gap:

- Include update in DISC DEXPI Requirements document
- Include example in Blueprint files

3.12 Prepare Symbols for Profile

Update symbol machine readable files to ensure correct information is made available to DISC DEXPI profile. Not all DISC DEXPI symbols have been drafted with correct sizing attribute-to-geometry consistency as required by the DISC DEXPI profile. Any inconsistent symbols are required to be updated.

Gap:

- Ensure each symbol is defined in a machine-readable format with correct:
 - sizing information and attribute-to-geometry consistency
 - origo placement
- Include updates in DISC DEXPI Profile

3.13 Update requirements to include DEXPI 2.0 conceptual model updates

A small number of name changes are introduced in DEXPI 2.0 that impact the DISC DEXPI requirement documentation.

Gap:

- Include updates in DISC DEXPI Requirements document

- Include updates to DISC DEXPI Profile

3.14 Creation of machine-readable DISC DEXPI Profile

The machine-readable profile (DiscProfile RC 1.xml) provides easy access to the content of the various sources (Excel file with several tables; Drawio and SVG files) that describe the DISC DEXPI Profile. While in principle machine readable, the so-far existing files provide only cumbersome access for importing tools.

From a formal point of view, the DISC DEXPI Profile

- [1] extends the existing DEXPI Information Model RC (Release Candidate) 1 with new classes and attributes and
- [2] imposes constraints on the usage of classes and attributes (both DEXPI- and DISC-defined), e.g., by prescribing the use of certain required attributes for certain classes ("FLO") and the use of predefined symbols to represent engineering objects such as equipment items and piping components.

The new DEXPI 2.0 Exchange Format (DEXPI XML) has been designed to support [1] extensions. For example, the following excerpt from DiscProfile RC 1.xml shows the definition of a new class type:

```
<ConcreteClass name="AcousticNoiseReducer" superTypes="Plant/Piping.PipingComponent">
  <Data property="DexpiMeta/rdl_uri">
    <String>http://noaka.org/rdl/AcousticNoiseReducer</String>
  </Data>
  <Data property="DexpiMeta/rdl_label">
    <String>ACOUSTIC NOISE REDUCER</String>
  </Data>
</ConcreteClass>
```

Another excerpt demonstrates the extension of the existing DEXPI class Equipment with a property called ObjectDisplayName of type String or Undefined:

```
<ClassExtension name="EquipmentExtension" baseType="Plant/Equipment.Equipment">
  <DataProperty name="ObjectDisplayName" isOrdered="false" isUnique="false" lower="0" upper="1">
    <UnionDataType>
      <DataTypeReference type="Builtin/Undefined"/>
      <DataTypeReference type="Builtin/String"/>
    </UnionDataType>
    <Data property="DexpiMeta/rdl_uri">
      <String>http://noaka.org/rdl/ObjectDisplayNameAssignmentClass</String>
    </Data>
    <Data property="DexpiMeta/rdl_label">
      <String>OBJECT DISPLAY NAME ASSIGNMENT CLASS</String>
    </Data>
  </DataProperty>
  ...
</ClassExtension>
```

DEXPI does not yet provide a canonical solution for [2] constraints. To this end, a proposal for a suitable DEXPI extension (Profile RC 1.xml, work in progress) is required. The current version of this extension supports the definition of graphical symbols with variants depending on object data (e.g., valve positions) as well as the assignment of these symbols to certain types of engineering objects.

The following excerpt defines DISC symbol ND0004 for DoubleBlockAndBleedValves. The symbol

exists in two variants, depending on the ValvePosition:

```
<Object name="ND0004" type="Profile/Symbol">
  <Components property="Variants">
    <Object name="ND0004_0" type="Profile/SymbolVariant">
      <Components property="Primitives">
        <Object type="Core/Diagram.PolyLine">...</Object>
        ...
      </Components>
      ...
    </Object>
    <Object name="ND0004_1" type="Profile/SymbolVariant">
      <Data property="Condition">
        <String>ValvePosition = 'NC'</String>
      </Data>
      <Components property="Primitives">
        <Object type="Core/Diagram.PolyLine">...</Object>
        ...
      </Components>
      ...
    </Object>
  </Components>
  <Data property="DexpiMeta/usage">
    <String>DiscProfile.InformationModel.DoubleBlockAndBleedValve</String>
  </Data>
</Object>
```

Further work will include the remaining information from the Excel, Drawio, and SVG sources, in particular

- piping and instrumentation node positions,
- label placement and displayed text, and
- manipulation constraints for symbols (rotation, scaling, ...).

For validation and demonstration of the machine-readable profile, a human-friendly HTML rendering (DISC Profile MS 1 HTML) has been produced to enable easy and navigable access to the profile.

- New DISC class: AcousticNoiseReducer

AcousticNoiseReducer

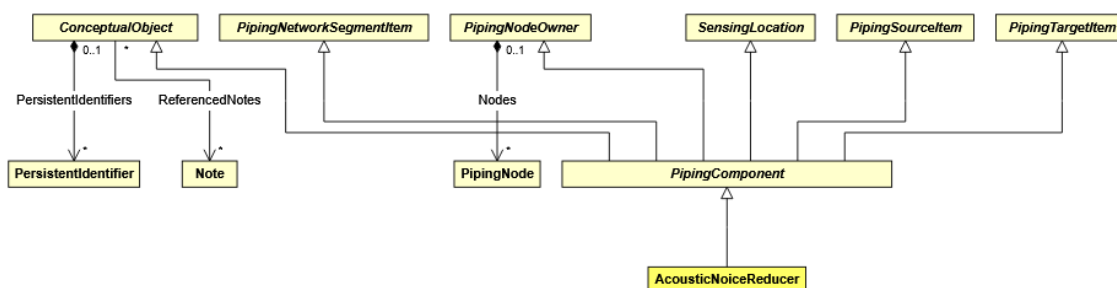
[DiscProfile::InformationModel::AcousticNoiseReducer](#)

Overview

Concrete Class

RDL: ACOUSTIC NOISE REDUCER (<http://noaka.org/rdl/AcousticNoiseReducer>)

(Fullscreen Diagram)



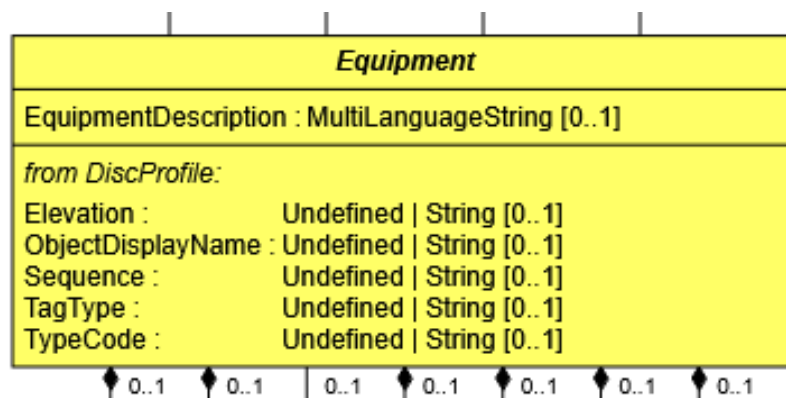
Supertypes

- [PipingComponent](#)

Example in DEXPI XML

```
<Object type="DiscProfile/InformationModel.AcousticNoiseReducer" />
```

- Extension of Equipment class



- Symbol ND0004

ND0004

DiscProfile::SymbolCatalogue::ND0004

Overview

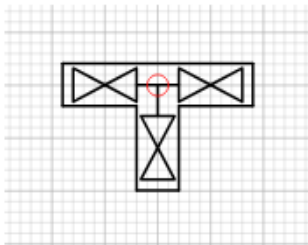
Object (Symbol)

Usage

DoubleBlockAndBleedValve

Default

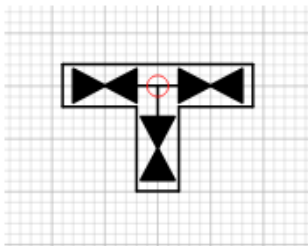
(Fullscreen Diagram)



Variant 1

Condition: ValvePosition = 'NC'

(Fullscreen Diagram)



Gap:

- Proposal for a suitable DEXPI extension (Profile RC 1.xml, work in progress) to cover [2] constraints
- Extraction of remaining information from the Excel, Drawio, and SVG sources and inclusion in DiscProfile RC 1

3.15 Blueprint Files

Already in the NOAKA DEXPI Project, a series of blueprint files has been created. Each file is a DEXPI Proteus XML file with a NOAKA-compliant variant of the DEXPI reference P&ID. Within the series of files, small changes, such as the addition of a valve or the removal of an instrument, mimic typical workflows that CAE tools have to support.

Gap:

- The existing blueprint files have to be converted to DEXPI 2.0 (new data format; refactoring of “custom” classes and attributes in DEXPI 1.3/1.4 to equivalent DEXPI 2.0 constructs).
- New blueprint files have to be created that cover some of the new requirements (cf. gaps in domain modelling above).

3.16 Validation Method

A method for the validation of DEXPI 2.0 files created by the involved CAE tools is required.

Validation should cover

- formal correctness (compliance with DEXPI XML format, DEXPI information model, and DISC DEXPI profile) and
- correctness and completeness in terms of content (e.g., while reproducing a blueprint file, the conceptual information (objects and attributes) should be both correct and complete in a strict sense; for the graphics, certain deviations are admissible provided an engineer perceives the diagram as equivalent to the original blueprint’s graphics).

Gap:

- scoring system
 - weighting
-

4 OUT-OF-SCOPE GAPS

- Engage with PCA to take ownership of the DISC Profile.
- Work with PCA initiative to take ownership of NCS Symbol Legend.
- Work with company drafting standards to enhance the foundation for mapping towards DEXPI.